

Synoptic CB: Porewater Nutrients

September 2025 Samples

2026-08-21

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##Run Information

```
cat("Run Information: NAME ") #lets you know what section you're in
```

Run Information: NAME

#set the run date & user name

```
run_date <- "06/30/2026"  
sample_year <- "2025"  
sample_month <- "SEPTEMBER"  
user <- "Stephanie Wilson"
```

##Data entered incorrectly

```
Old_ID_1 <- c("MSM_202509_UP_LysC_45cm ")  
New_ID_1 <- c("MSM_202509_UP_LysC_45cm")
```

#identify the files you want to read in

#read in as a list to accommodate multiple runs in a month

```
NOx_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NOx_202509_1.csv",  
              "Raw Data/SEAL_COMPASS_Synoptic_NOx_202509_2.csv",  
              "Raw Data/SEAL_COMPASS_Synoptic_NOx_202509_3.csv")  
NH3_PO4_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202509_1.csv",  
                  "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202509_2.csv",  
                  "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202509_3.csv",  
                  "Raw Data/SEAL_COMPASS_Synoptic_PO4_202509_3_GOOD.csv")
```

Define the file path for QAQC log file - NO Need to change just check year

```
file_path <- "Raw Data/SEAL_COMPASS_Synoptic_QAQC_Log_2025.csv"  
final_path <- "Processed Data/COMPASS_Synoptic_Nutrients_202509-11.csv"
```

#record any notes about the run or anything other info here:

```
run_notes <- "On the last NH3/PO4 run the PO4 standard curve was bad and it seemed like reagents were
```

#Set up file path for metadata

```
#downloaded metadata csv - downloaded from Google drive as csv for this year  
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data & Clean

```
##Fix data entered incorrectly
```

```
####Fix Sample IDs####
```

```
df_all$Sample_Name[df_all$Sample_Name == Old_ID_1] <- New_ID_1
```

```
#move the date to the correct place in the ones entered incorrectly
```

```
df_all$Sample_Name <- sub(  
  "(GCW|GWI|MSM|SWH)_(TR|WC|UP|SW)_([0-9]{6})_(LysA|LysB|LysC)_(10|20|45|NA)cm$",  
  "\\1_\\3_\\2_\\4_\\5cm",  
  df_all$Sample_Name  
)
```

```
df_all$Sample_Name <- sub(  
  "(GCW|GWI|MSM|SWH)_(SW)_([0-9]{6})_(A|B|C)$",  
  "\\1_\\3_\\2_\\4",  
  df_all$Sample_Name  
)
```

0.2 Assessing standard Curves

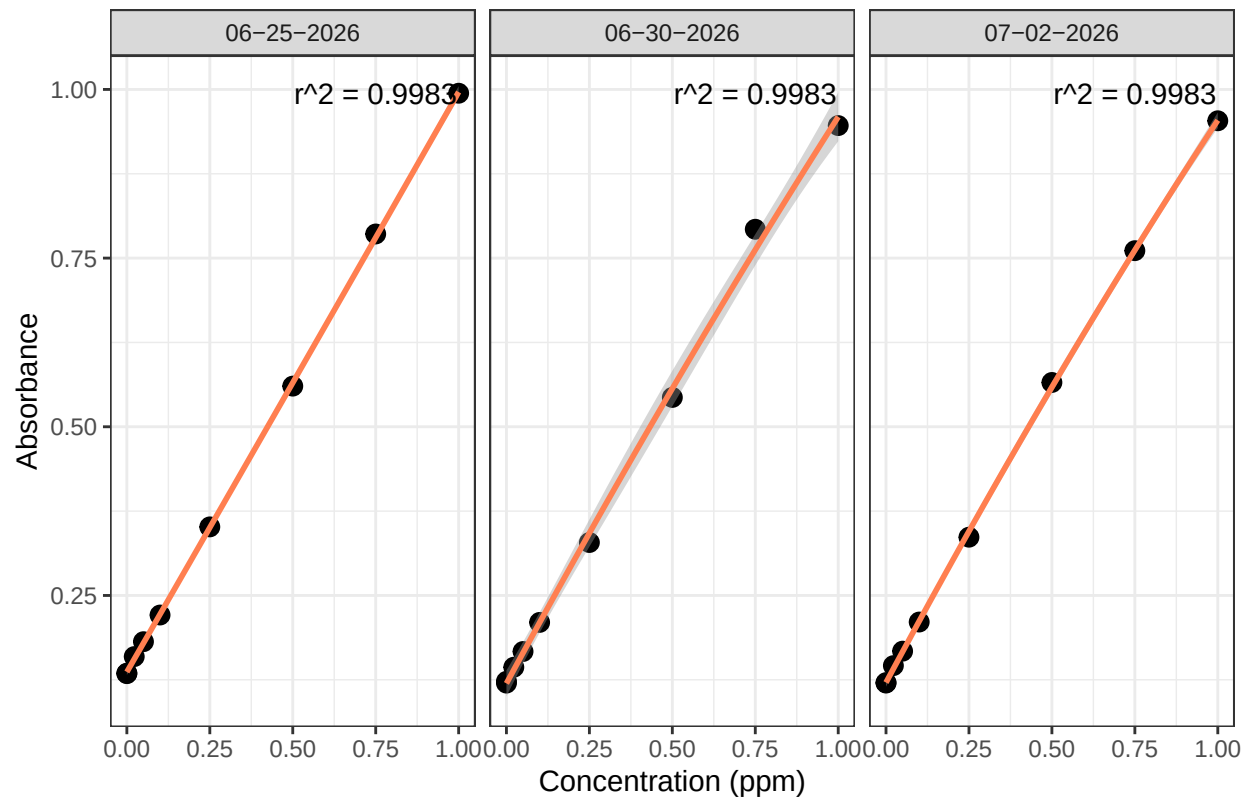
```
#Pull out standards data
```

```
## Assess Standard Curves
```

0.3 Plot standards data

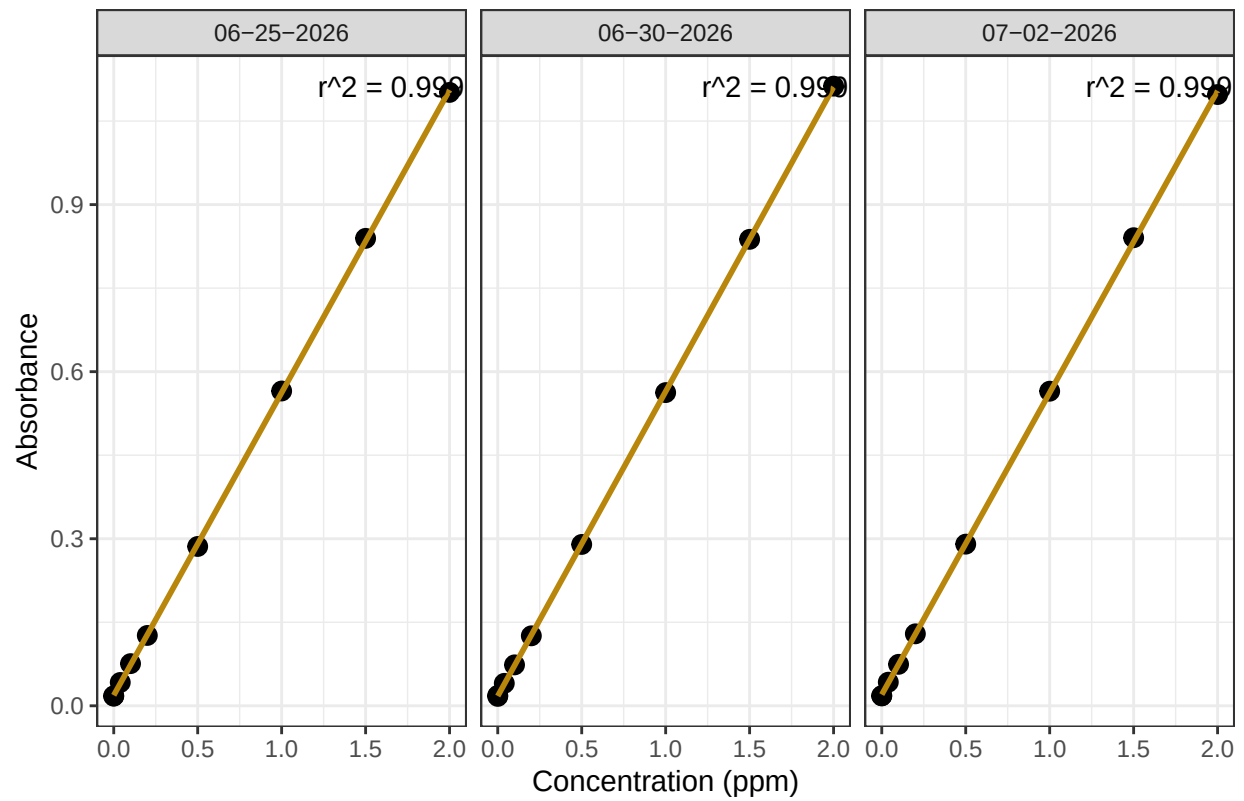
```
## Assess Standard Curves
```

NOx Standard Curve



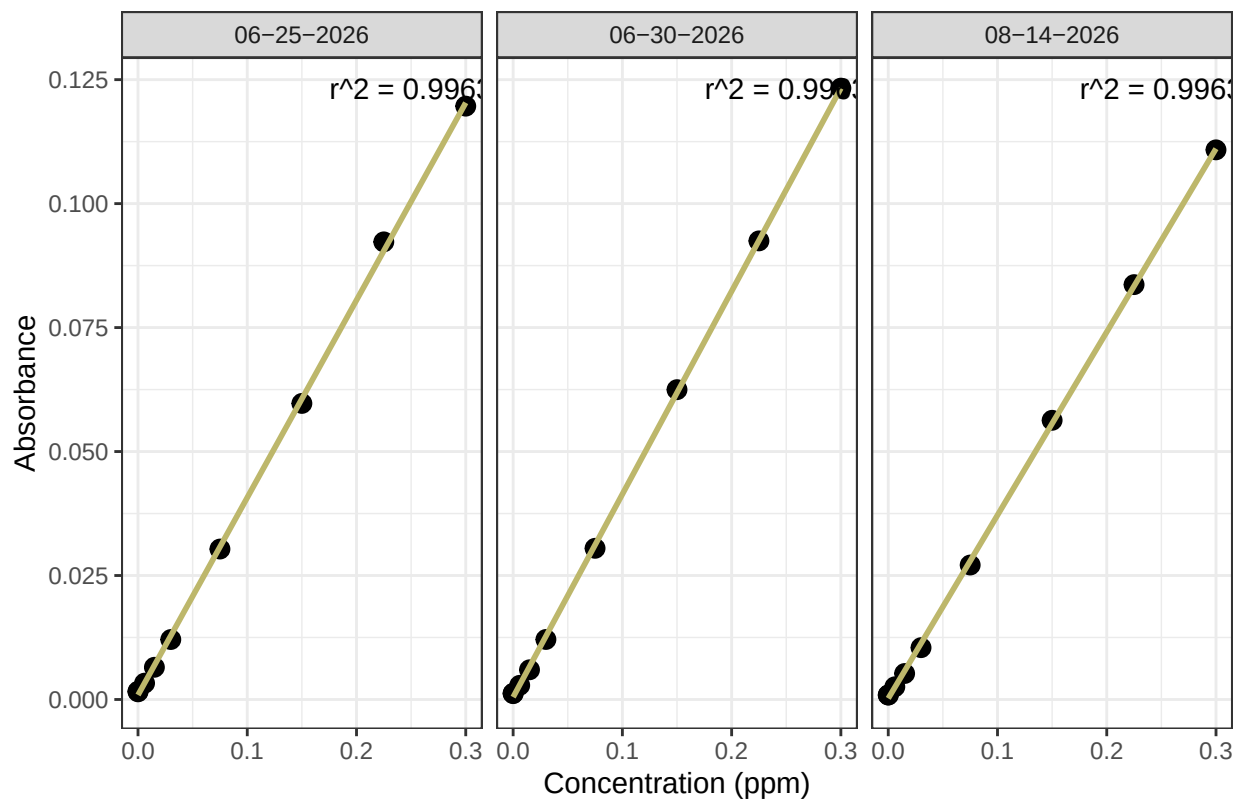
```
## 'geom_smooth()' using formula = 'y ~ x'
```

NH3 Standard Curve



```
## 'geom_smooth()' using formula = 'y ~ x'
```

PO4 Standard Curve



```
## [1] "NOx Curve r2 GOOD - PROCEED"
```

```
## [1] "NH3 Curve r2 GOOD - PROCEED"
```

```
## [1] "PO4 Curve r2 GOOD - PROCEED"
```

```
## [1] "QAQC log file exists and has been read into the code."
```

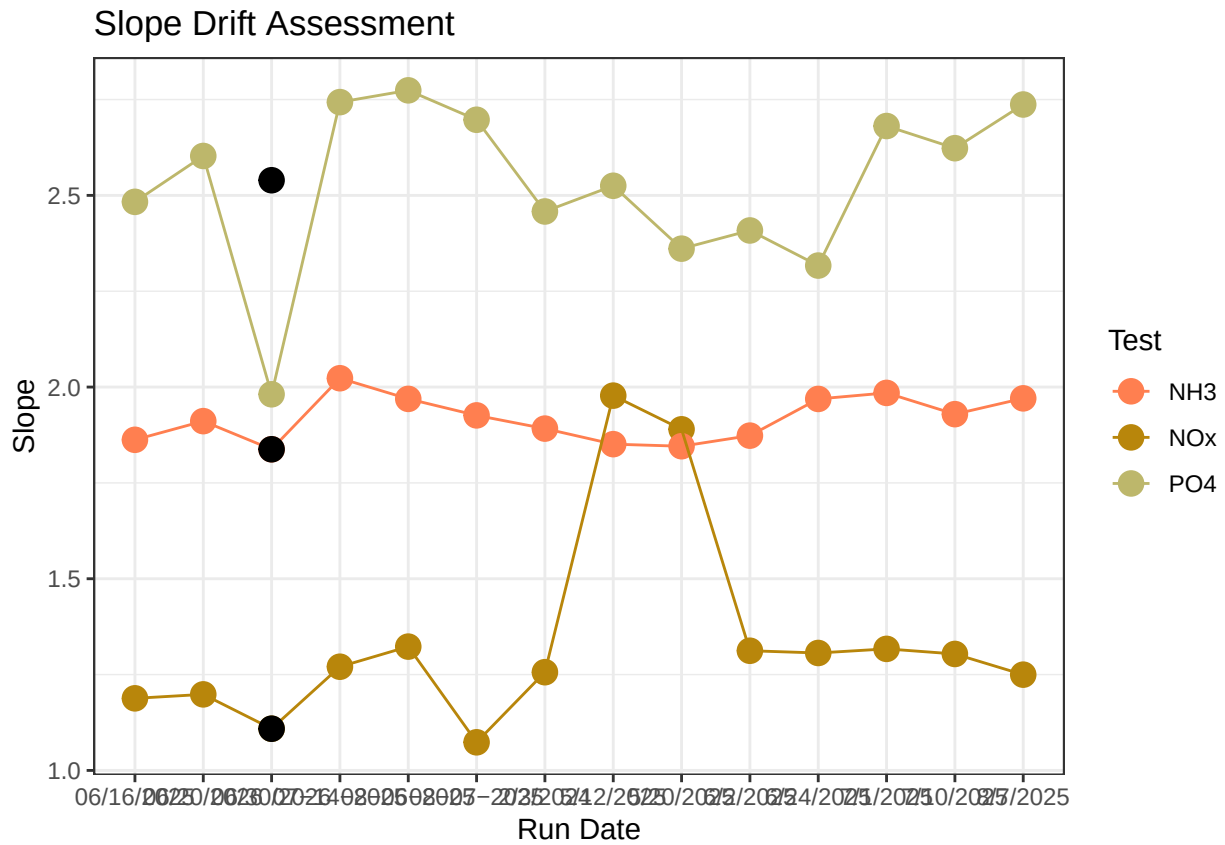


Table 1: Average Slope by Analyte

Test	avg_slope
NH3	1.917
NOx	1.341
PO4	2.528

0.4 Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## Duplicated samples: GCW_202509_WC_LysA_45cm, GCW_202509_WC_LysB_10cm, GCW_202509_WC_LysB_20cm, GCW_202509_WC_LysB_45cm
```

```
##
```

```
## All duplicated samples have valid dilutions. No naming issues detected.
```

0.5 Performance Check

```
## [1] "NOx pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 1.550669
```

```
## Expected = 1.51
```

```
## [1] "NH3 pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 1.097768
```

```
## Expected = 1.034
```

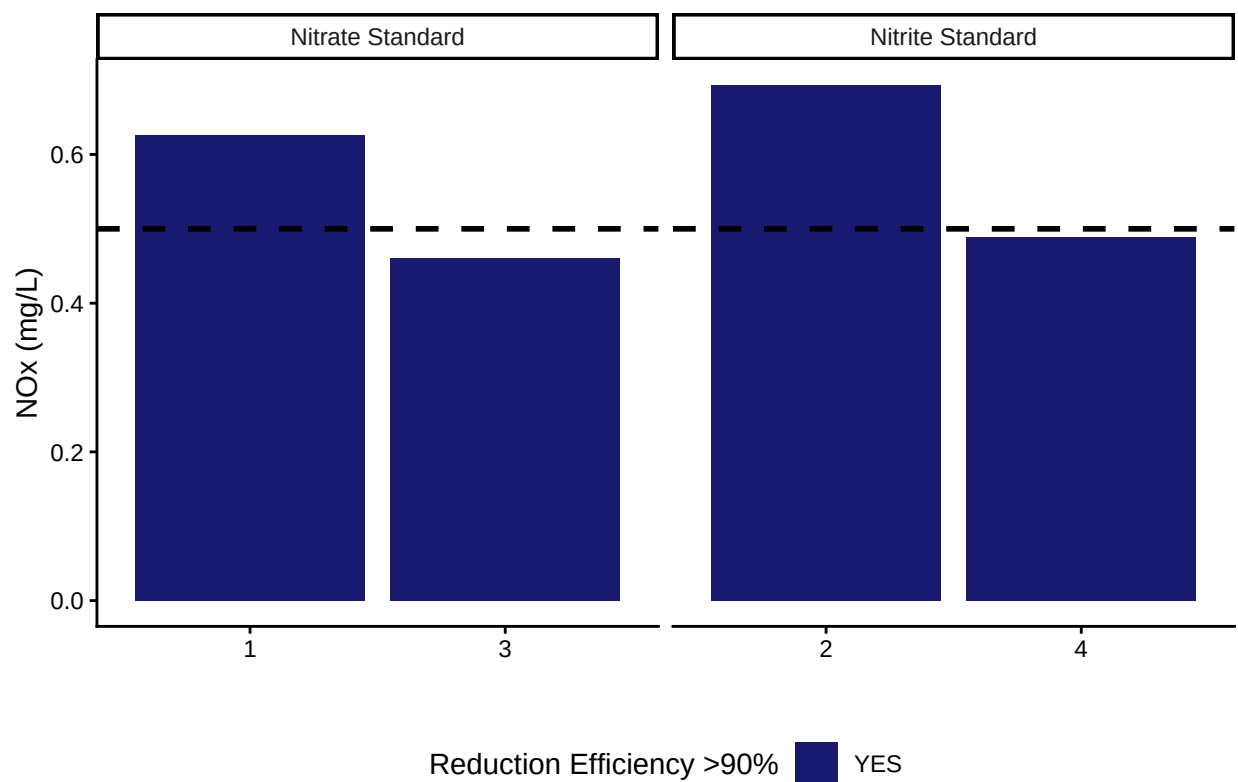
```
## [1] "P04 pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 0.8710093
```

```
## Expected = 0.824
```

```
##Check NOx Reduction Efficiency
```

```
## Assess Reduction Efficiency
```

```
## [1] "Mean NOx Reduction Efficiency >95% - PROCEED"
```

```
## [1] 113.4839
```

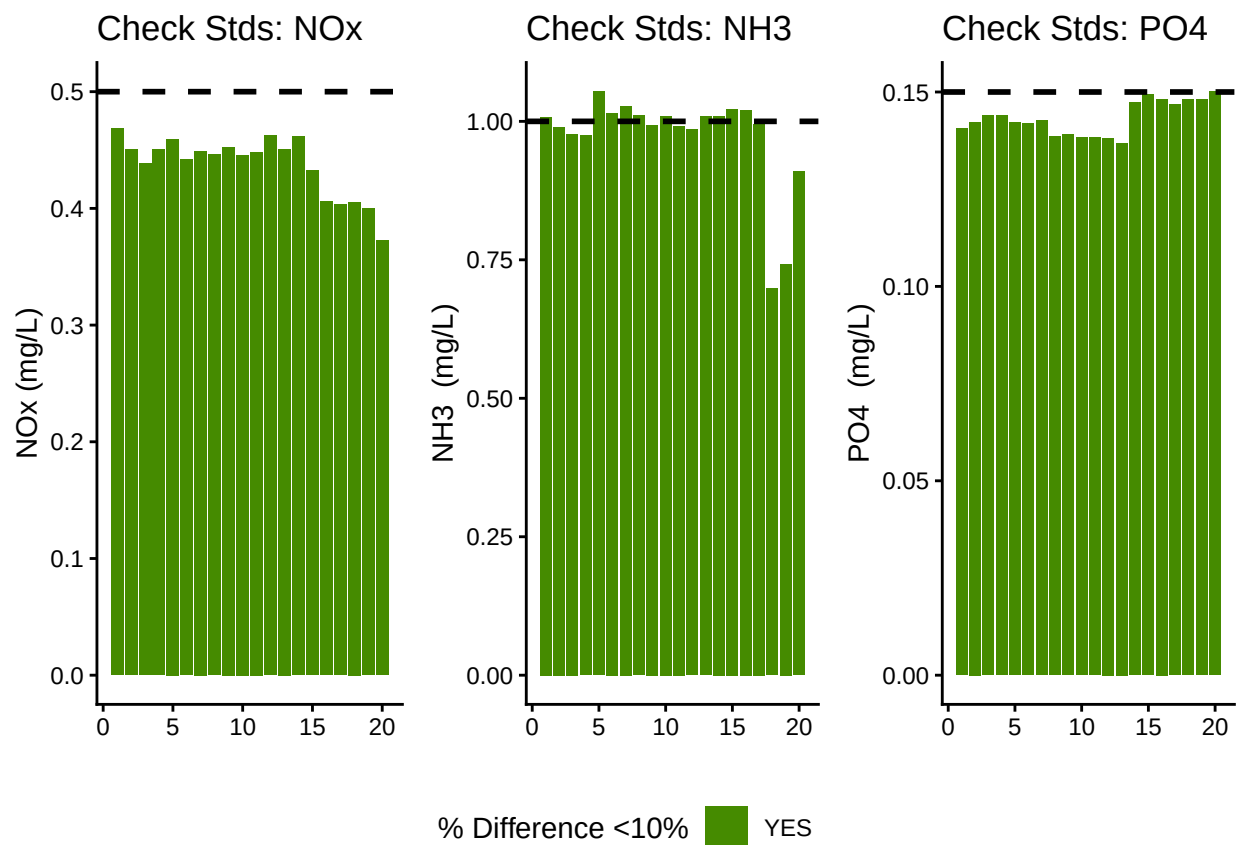
0.6 Analyze the Check Standards

```
## Analyze Check Standards
```

```
## [1] "NOx Check Standard RSD within Range - PROCEED"
```

```
## [1] "NH3 Check Standard RSD within Range - PROCEED"
```

```
## [1] "PO4 Check Standard RSD within Range - PROCEED"
```



```
## [1] ">60% of NOx Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of NH3 Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of PO4 Check Standards are within range of expected concentration - PROCEED"
```

0.7 Analyze Blanks

```
## Assess Blanks
```

```
## [1] ">60% of NOx Blanks are below the lower 25% quartile of samples or 1/2 detection limit - PROCEED"
```

```
## [1] ">60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED"
```

```
## [1] ">60% of PO4 Blanks are below the lower 25% quartile of samples- PROCEED"
```

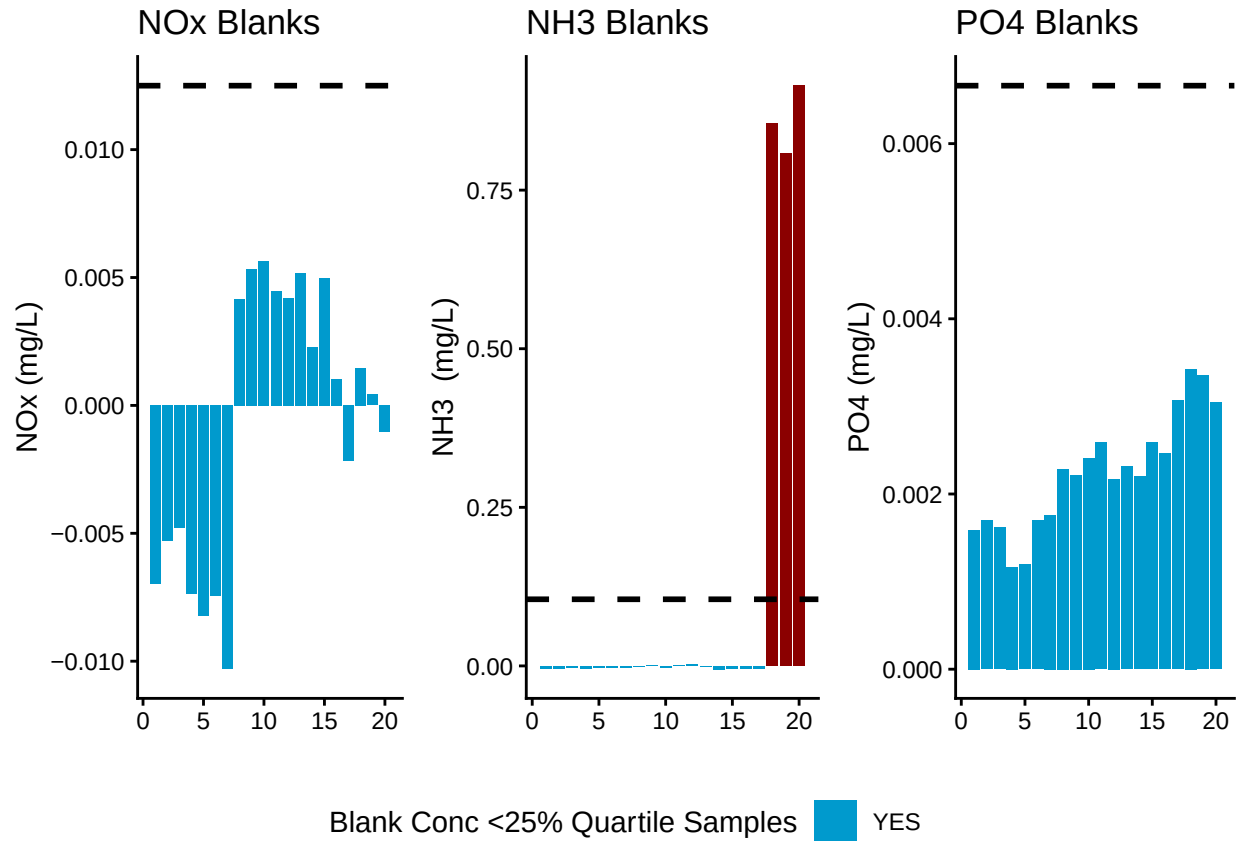


Table 2: Mean Concentration of Blanks

Test	Blank_Mean_Conc
NOx	-0.0007
NH3	0.1269
PO4	0.0022

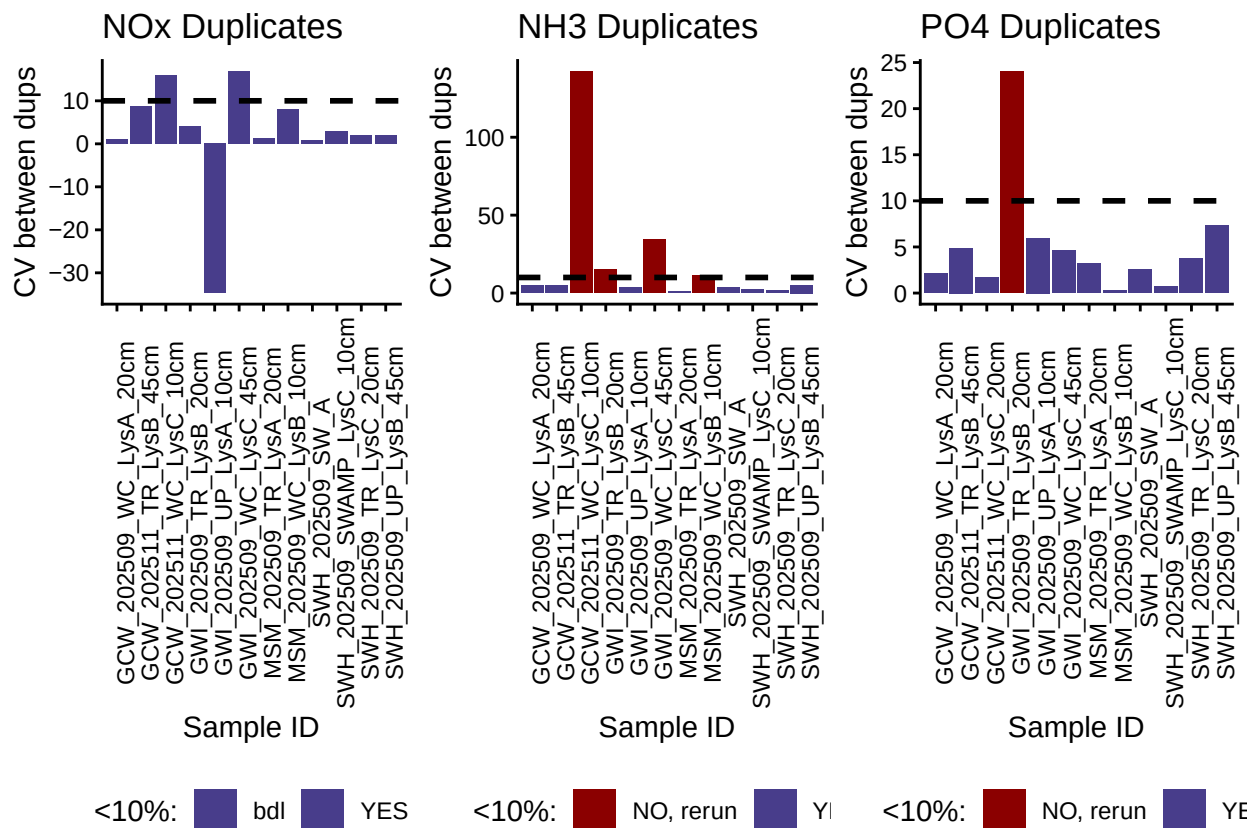
0.8 Analyze Duplicates

Analyze Duplicates

[1] ">60% of NOx Duplicates have a CV <10% - PROCEED"

[1] ">60% of NH3 Duplicates have a CV <10% - PROCEED"

[1] ">60% of PO4 Duplicates have a CV <10% - PROCEED"



0.9 Spikes

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

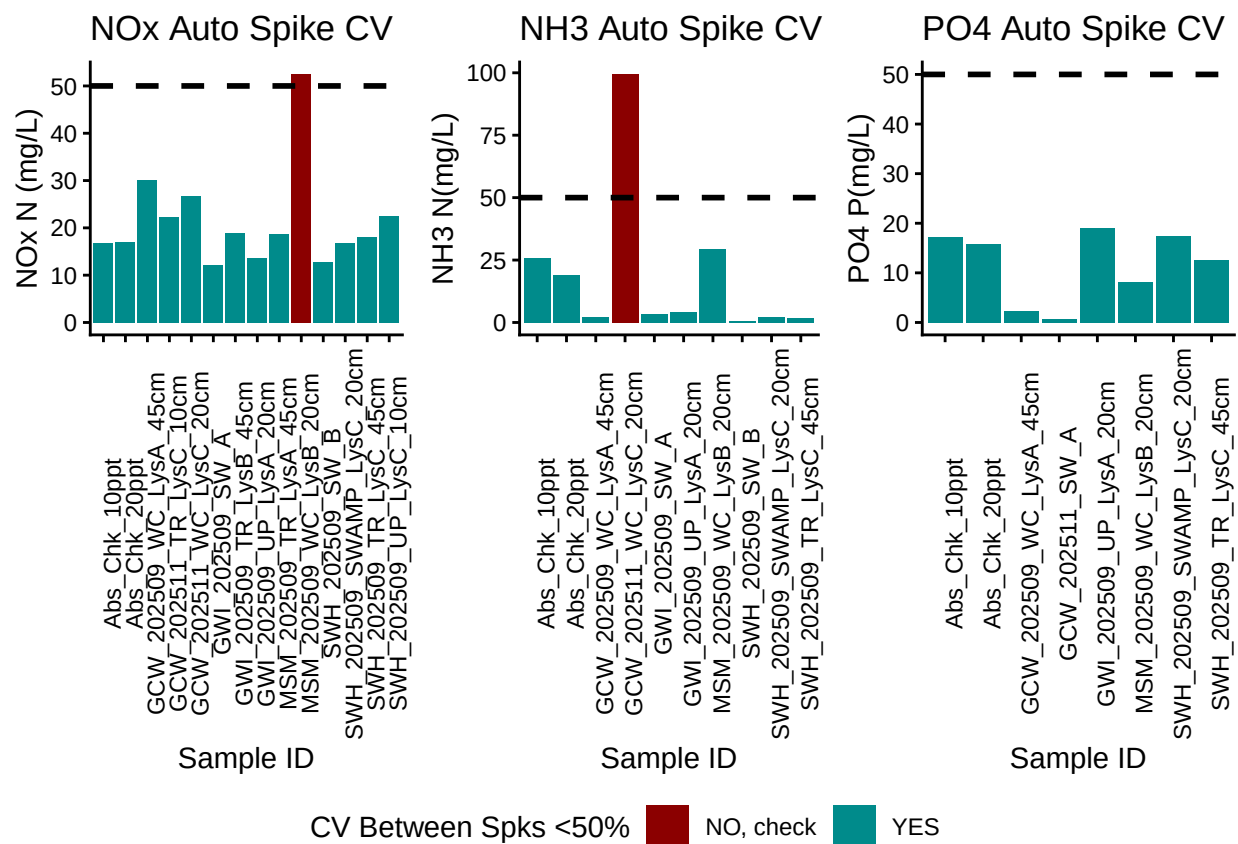
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
```

```
## i Please use 'linewidth' instead.
```

```
## This warning is displayed once per session.
```

```
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

```
## generated.
```



0.10 Matrix Effects

```
## [1] "NO NOx Matrix Effect, PROCEED"
```

```
## [1] "NO NH3 Matrix Effect, PROCEED"
```

```
## [1] "NO PO4 Matrix Effect, PROCEED"
```

0.11 Unit Converted Data Column Added (mg/L to uM)

0.12 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

0.13 Pull out sample id information

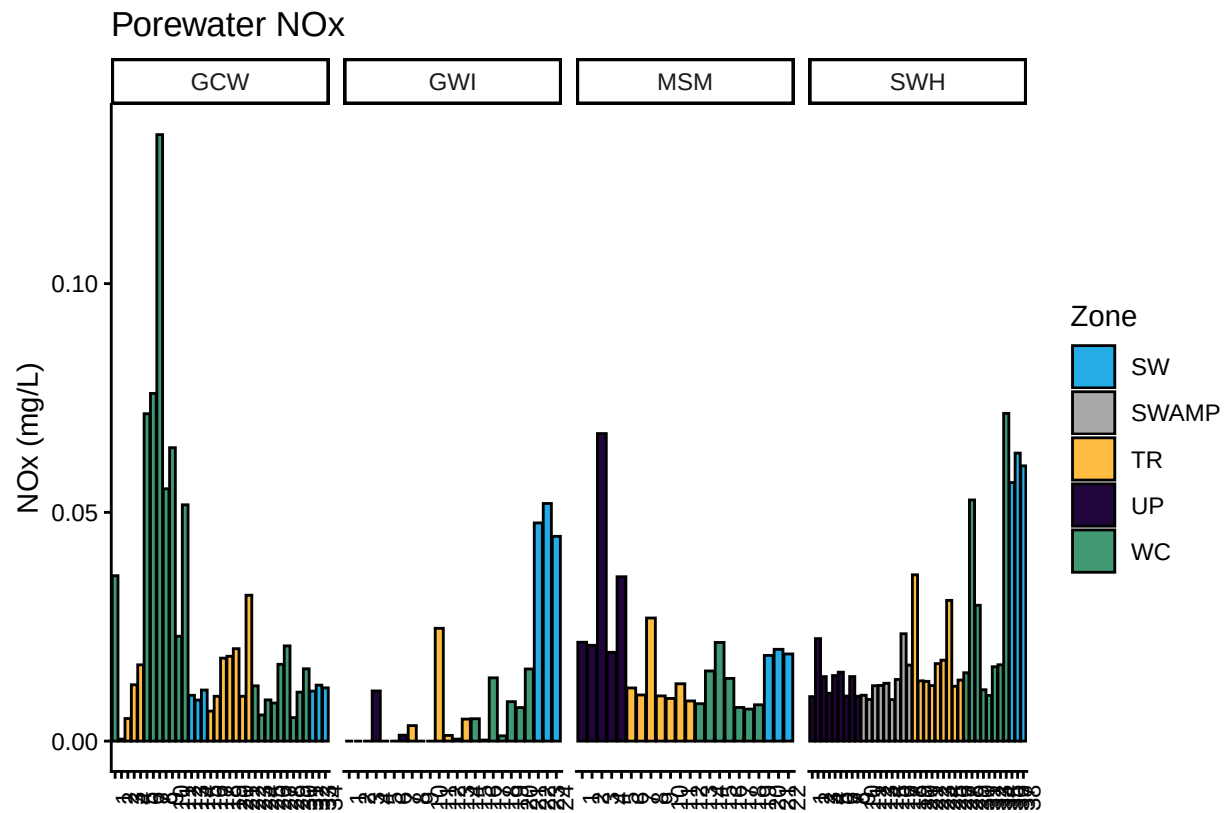
```
## Sample Processing
```

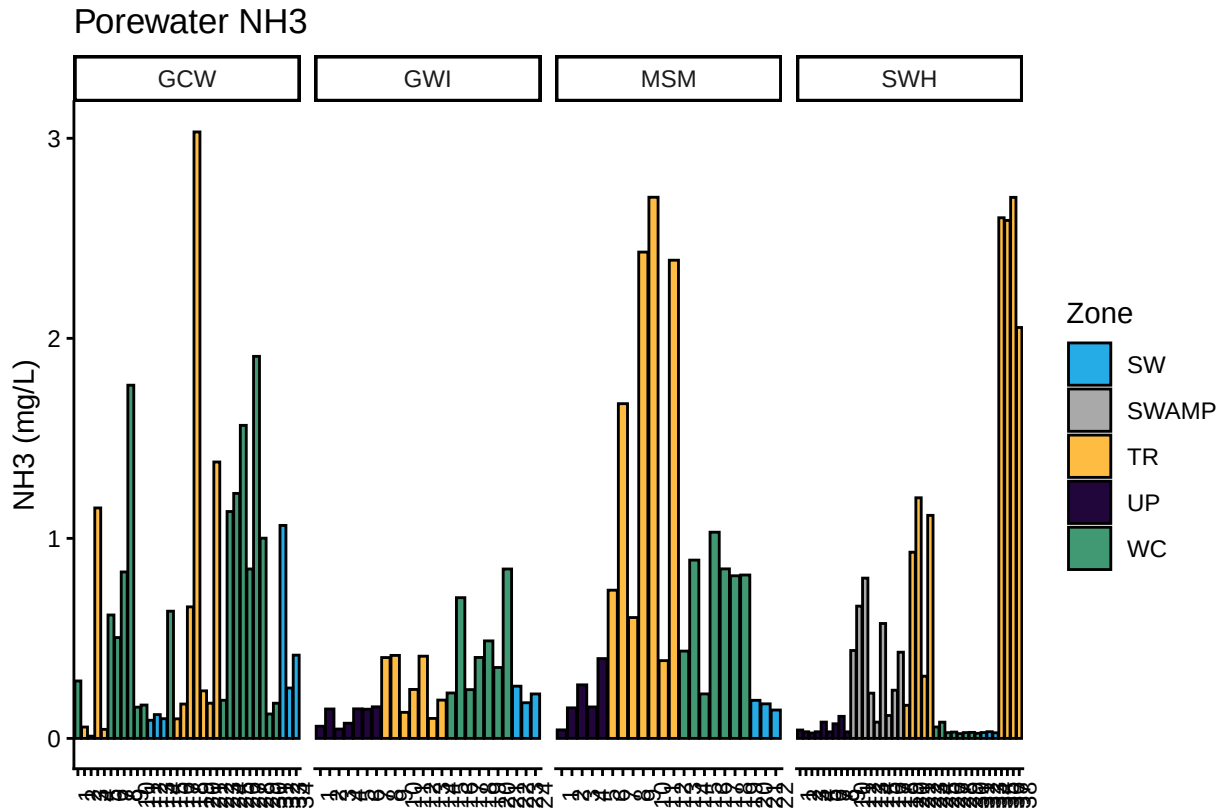
0.14 Check to see if samples run match metadata & merge info

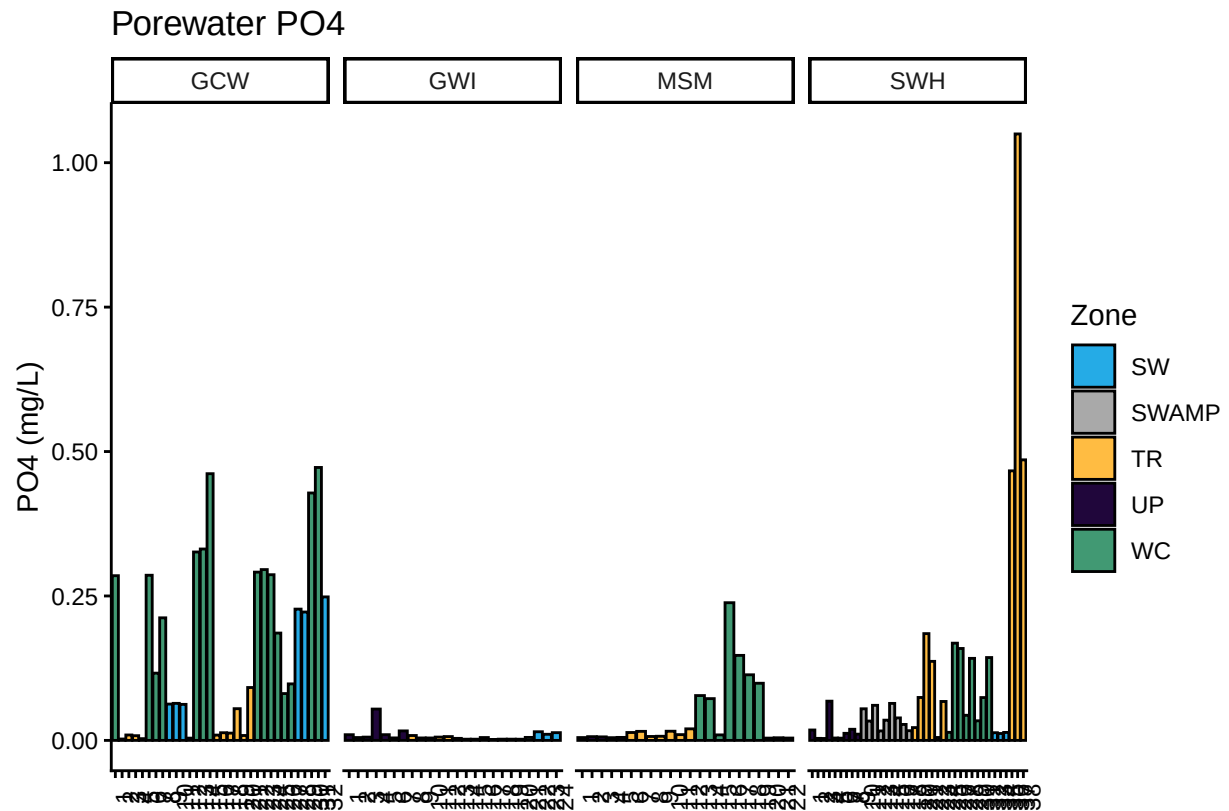
```
## Check Sample IDs with Metadata
```

```
## All sample IDs are present in metadata.
```

Visualize Data







0.16 Export Processed Data

#end